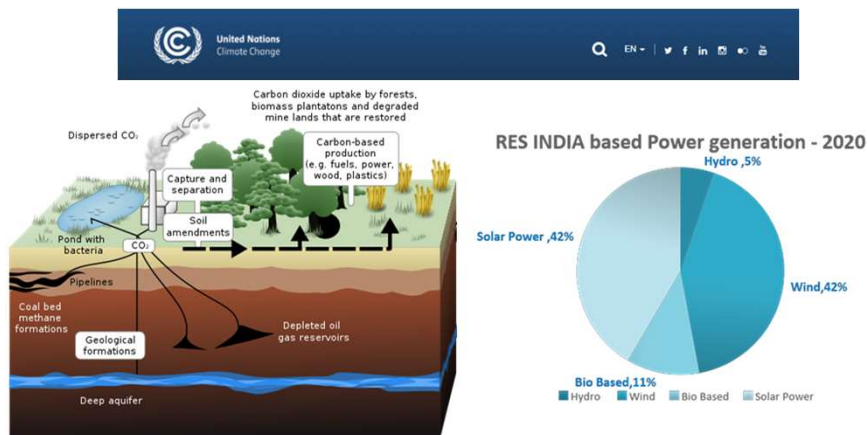


CH416 – CO₂ Capture and Utilization

Standard Policies



Department of Chemical Engineering

National Institute of Technology, Warangal

30-07-2025

CH447 – CO₂ Capture, Sequestration & Utilization

1

UNFCCC – Annex I Countries

The screenshot shows the UNFCCC website interface. The header includes the United Nations Climate Change logo and navigation links. The main content area is titled 'UNFCCC Process - Parties'. It includes a table of countries categorized by 'Type of Party to the Convention' (Annex I and Non-Annex I) and 'Also Party to' (Kyoto Protocol and Paris Agreement). The table lists 40 countries, including Australia, Austria, Belarus, Belgium, Bulgaria, Canada, Croatia, Cyprus, Czechia, Denmark, Estonia, European Union, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Monaco, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Russian Federation, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, and United States of America. A sidebar on the right shows the 'STATUS OF RATIFICATION' for the UNFCCC Convention, Kyoto Protocol, Doha Amendment to the Kyoto Protocol, and Paris Agreement.

United Nations
Framework
Convention on
Climate Change

Kyoto Protocol –

Targeted to achieve the stabilization of green house gases (GHG) concentrations in the atmosphere

Annex-I Countries to reduce by an average of 5.2% below 1990 levels

Inventory levels of National GHGs

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2

Carbon Sequestration Leadership Forum (CSLF)



"To facilitate the development of improved cost-effective technologies for the separation and capture of carbon dioxide for its transport and long-term safe storage; to make these technologies broadly available internationally; and to identify and address wider issues relating to carbon capture and storage. This could include promoting the appropriate technical, political, and regulatory environments for the development of such technology."

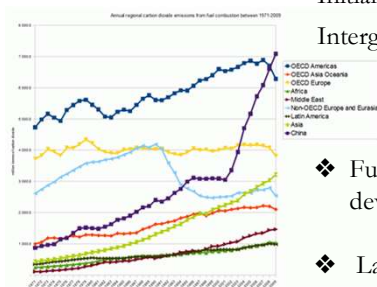
Initial estimation of working time of around 10 years

Intergovernmental Panel on Climate Change (IPCC)

- Continuous increase of carbon emissions
- Only minor projects were funded under this

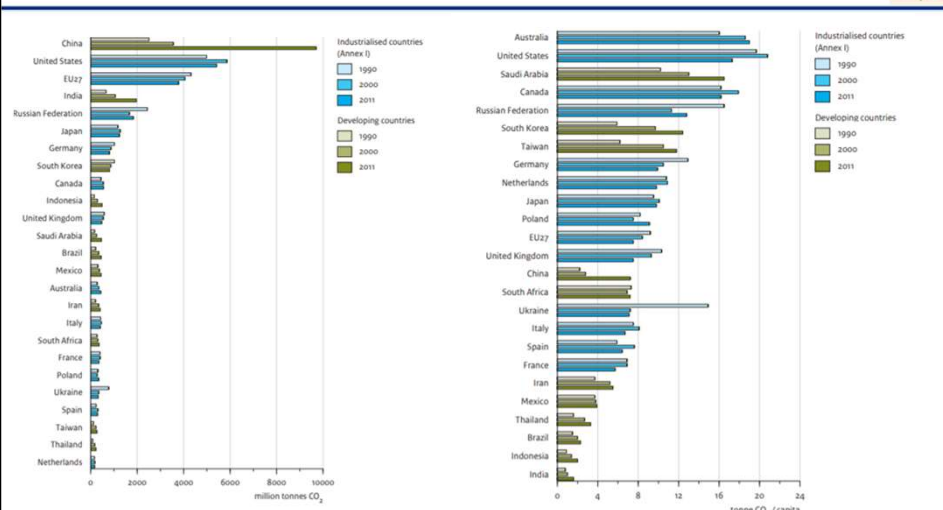
❖ Funds based on the per-capita emission is rejected by developed countries.

❖ Later this one also failed because of funding and serious implementations.



https://commons.wikimedia.org/wiki/File:Yearly_trends_in_annual_regional_carbon_dioxide_emissions_from_fuel_combustion_between_1971_and_2009.png

Global Emissions – CO₂ trends

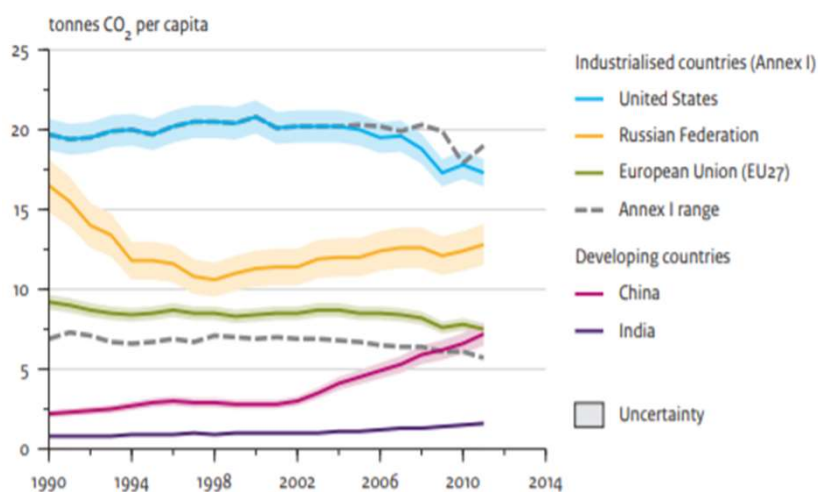


CO₂ emission **Per Country** from fossil fuel usage and cement production

https://www.pbl.nl/sites/default/files/downloads/PBL_2012_Trends_in_global_CO2_emissions_500114022.pdf

CO₂ emission **Per Capita** from fossil fuel usage and cement production

Global Emissions – CO₂ trends



https://www.pbl.nl/sites/default/files/downloads/PBL_2012_Trends_in_global_CO2_emissions_500114022.pdf

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CH447 – CO₂ Capture, Sequestration & Utilization

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Global Emissions – CO₂ trends



CO₂ emissions in 2011 (million tonnes CO₂) and CO₂/capita emissions, 1990–2011 (tonne CO₂/person)

Country	Emissions 2011	Per capita emissions				Change 1990–2011	Change 1990–2011 in %	Change in CO ₂ 1990–2011 in %	Change in population 1990–2011, in %
		1990	2000	2010	2011				
Annex I*									
United States	5420	19.7	20.8	17.8	17.3	-2.4	-12%	9%	19%
EU27	3790	9.2	8.4	7.8	7.5	-1.7	-18%	-12%	6%
Germany	810	12.9	10.5	10.2	9.9	-3	-23%	-21%	4%
United Kingdom	470	10.3	9.3	8.1	7.5	-2.8	-27%	-20%	8%
Italy	410	7.5	8.1	6.9	6.7	-0.8	-11%	-4%	7%
France	360	6.9	6.9	6.1	5.7	-1.2	-17%	-9%	10%
Poland	350	8.2	7.5	8.8	9.1	0.9	11%	11%	1%
Spain	300	5.9	7.6	6.3	6.4	0.5	8%	25%	16%
Netherlands	160	10.8	10.9	10.5	9.8	-1	-9%	-2%	11%
Russian Federation	1830	16.5	11.3	12.4	12.8	-3.7	-22%	-25%	-4%
Japan	1240	9.5	10.1	10	9.8	0.3	3%	2%	7%
Canada	560	16.2	17.9	16	16.2	0	0%	74%	19%
Australia	430	16.0	18.6	17.9	19.0	3	19%	57%	24%
Ukraine	320	14.9	7.2	6.7	7.1	-7.8	-52%	-58%	-12%
Non Annex I									
China	9700	2.2	2.8	6.6	7.2	5	227%	287%	15%
India	1970	0.8	1.0	1.5	1.6	0.8	100%	198%	30%
South Korea	610	5.9	9.7	12.2	12.4	6.5	110%	141%	11%
Indonesia	490	0.9	1.4	2	2.0	1.1	122%	210%	24%
Saudi Arabia	460	10.2	13.0	15.8	16.5	6.3	62%	181%	43%
Brazil	450	1.5	2.0	2.2	2.3	0.8	53%	106%	24%
Mexico	450	3.7	3.8	3.9	3.9	0.2	5%	45%	27%
Iran	410	3.7	5.2	5.4	5.5	1.8	49%	100%	27%
South Africa	360	7.3	6.9	7.1	7.2	-0.1	-1%	35%	27%
Taiwan	270	6.2	10.5	11.7	11.8	5.6	90%	119%	13%
Thailand	230	1.6	2.7	3.3	3.3	1.7	106%	155%	18%

Source of population data: UNPD, 2010 (WPP Rev. 2010)

* Annex I countries: industrialised countries with annual reporting obligations under the UN Framework Convention on Climate Change (UNFCCC) and emission targets under the Kyoto Protocol. The United States has signed but not ratified the protocol, and thus the US emission target in the protocol has no legal status.

➤ USA, EU, Germany, UK etc. reduced the per capita emission during 20 year block.

https://www.pbl.nl/sites/default/files/downloads/PBL_2012_Trends_in_global_CO2_emissions_500114022.pdf

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CH447 – CO₂ Capture, Sequestration & Utilization

8

INDIA – Energy Needs



BP Energy Outlook – 2019

Insights from the Evolving transition scenario – India



BP Energy Outlook – 2019

Insights from the Evolving transition scenario – India



India's share of total global primary energy demand is set to roughly double to ~11% by 2040, underpinned by strong population growth and economic development

1. India accounts for more than a quarter of net global primary energy demand growth between 2017-2040.
2. 42% of this new energy demand is met through coal, meaning CO₂ emissions roughly double by 2040.
3. Gas production grows but fails to keep pace with demand, implying a significant growth in gas imports.

+156%

Growth in India's primary energy consumption

11%

India's share of global energy consumption in 2040

58%

Share of coal in India's power generation mix

+116%

Growth in India's net CO₂ emissions by 2040

India's population increases by more than 350m, and the economy nearly doubles in size, meaning income per capita roughly doubles.

This growth in absolute terms means India's share of global primary energy demand jumps from 6% today to 11% by 2040.

Power generation increases by 207% to 4,761 TWh by 2040, accounting for 61% of primary energy demand growth.

Industry is the strongest source of final energy demand growth (+238 Mtoe followed by transport (+144 Mtoe) and non-combustible (+84 Mtoe).

Coal meets ~42% of India's new energy demand, increasing by 493 Mtoe. The majority (84%) of this additional consumption is met through domestic production.

Renewable energy consumption surges from ~20 Mtoe today to ~350 Mtoe by 2040 – concentrated mainly in the power sector and driven largely by growth in solar capacity. Yet despite the growth in renewables, coal continues to dominate India's power generation mix, accounting for 58% of output by 2040.

As a result, although the carbon intensity of India's power grid declines by 28% by 2040, it remains 58% above the global average.

India's total net CO₂ emissions roughly double to 500 by 2040, meaning India's share of global emissions increases from 7% today to 14% by 2040.

Although gas production increases modestly to ~75 Bcm, demand surges some 240% to reach 185 Bcm by 2040, meaning India's reliance on gas imports is set to continue to grow significantly.

Nuclear capacity continues to grow slowly and accounts for 4% of total power generation by 2040.

* Compound annual growth rate
† Oil supply includes crude oil, shale oil, oil sands, natural gas liquids, liquid fuels derived from coal and gas, and refinery gases, but excludes biofuels. Oil demand includes consumption of all liquid hydrocarbons but excludes biofuels.
‡ Includes electricity and the associated conversion losses in power generation.



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<https://www.bp.com/content/dam/bp/business-sites/en/global/corporate/pdfs/energy-economics/energy-outlook/bp-energy-outlook-2019-country-insight-india.pdf>

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CH447 – CO₂ Capture, Sequestration & Utilization

9

INDIA – Thermal Plants Installed Capacity



ALL INDIA INSTALLED CAPACITY (IN MW) OF POWER STATIONS (As on 31.08.2020) (UTILITIES)

Region	Ownership/ Sector	Mode wise breakup							Grand Total	
		Thermal				Nuclear	Hydro	RES * (MNRE)		
Coal	Lignite	Gas	Diesel	Total						
Northern Region	State	16659.00	250.00	2879.20	0.00	19788.20	0.00	5777.25	725.01	26290.46
	Private	22425.83	1080.00	558.00	0.00	24063.83	0.00	2817.00	16242.89	43123.72
	Central	14354.96	250.00	2344.06	0.00	16949.02	1620.00	11450.52	379.00	30398.54
	Sub Total	53439.79	1580.00	5781.26	0.00	60801.05	1620.00	20044.77	17346.90	99812.72
Western Region	State	21740.00	1040.00	2849.82	0.00	25629.82	0.00	5446.50	555.54	31631.86
	Private	32847.17	500.00	4676.00	0.00	38023.17	0.00	481.00	25397.44	63901.61
	Central	19147.95	0.00	3280.67	0.00	22428.62	1840.00	1627.50	666.30	26562.42
	Sub Total	73735.12	1540.00	10806.49	0.00	86081.61	1840.00	7555.00	26619.28	122095.89
Southern Region	State	19782.50	0.00	791.98	159.96	20734.44	0.00	11774.83	586.88	33096.15
	Private	12747.00	250.00	5340.24	273.70	18610.95	0.00	0.00	41755.25	60366.20
	Central	11835.02	2990.00	359.58	0.00	15184.60	3320.00	0.00	541.90	19046.50
	Sub Total	44364.52	3240.00	6491.80	433.66	54529.99	3320.00	11774.83	42884.03	112508.85
Eastern Region	State	7450.00	0.00	100.00	0.00	7550.00	0.00	3537.92	275.11	11363.03
	Private	6153.00	0.00	0.00	0.00	6153.00	0.00	96.00	1255.50	7504.50
	Central	13682.05	0.00	0.00	0.00	13682.05	0.00	1005.20	10.00	14697.25
	Sub Total	27285.05	0.00	100.00	0.00	27385.05	0.00	4639.12	1540.61	33564.78
North Eastern Region	State	0.00	0.00	498.86	36.00	534.86	0.00	422.00	233.25	1190.10
	Private	0.00	0.00	24.50	0.00	24.50	0.00	0.00	104.14	128.64
	Central	770.02	0.00	1253.60	0.00	2023.62	0.00	1263.50	30.00	3317.12
	Sub Total	770.02	0.00	1776.96	36.00	2582.98	0.00	1685.50	367.39	4635.86
Islands	State	0.00	0.00	0.00	40.05	40.05	0.00	0.00	5.25	45.30
	Private	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.87	24.87
	Central	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	5.10
	Sub Total	0.00	0.00	0.00	40.05	40.05	0.00	0.00	35.22	75.27
ALL INDIA	State	65631.50	1290.00	7119.86	236.01	74277.36	0.00	26958.50	2381.03	103616.90
	Private	74173.00	1830.00	10598.74	273.70	86875.45	0.00	3394.00	84780.09	175049.54
	Central	59790.00	3240.00	7237.91	0.00	70267.91	6780.00	15346.72	1632.30	94026.93
	Total	199594.50	6360.00	24956.51	509.71	231420.72	6780.00	45699.22	88793.43	372693.36

Figures at decimal may not tally due to rounding off

http://www.cea.nic.in/reports/monthly/installedcapacity/2020/installed_capacity-08.pdf

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CH447 – CO₂ Capture, Sequestration & Utilization

10

INDIA – Thermal Plants Installed Capacity



A capacity addition target of 88,537 MW excluding 30,000 MW of RES, was fixed for 12th Plan period. In the Twelfth Plan, a cumulative capacity addition of 99,209.6 MW was achieved. **This is the first time in the history of Five year Plans that the Capacity Addition target has been overachieved.** The details are as under:

(In MW)

Sector	Thermal		Hydro		Nuclear		Total	
	Target	Ach.	Target	Ach.	Target	Ach.	Target	Ach.
Central	14,878	15,868.6	6,004	2584	5,300	2,000	26,182	20,452.7
State	13,922	22,201.4	1,608	2276	0	0	15,530	24,477.4
Private	43,540	53,660.5	3,285	619	0	0	46,825	54,279.5
All India	72,340	91,730.5	10,897	5479	5,300	2,000	88,537	99,209.6

Capacity addition target and achievement for 2017-18:

For the year 2017-18 a capacity addition target of 13,171 MW including 500 MW of nuclear power was fixed. Against this, a capacity addition of 9,505 MW was achieved. The Sector-wise / Fuel-wise break up is as under:

(In MW)

	Thermal		Hydro		Nuclear		Total	
	Target	Ach.	Target	Ach.	Target	Ach.	Target	Ach.
Central	4880	3170	800	390	500	0	6180	3560
State	3546	1760	300	200	-	-	3846	1960
Private	2940	3780	205	205	-	-	3145	3985
Total	11366	8710	1305	795	500	0	13171	9505

https://powermin.nic.in/sites/default/files/uploads/overview_of_Capacity_Addition_during_12th_Plan.pdf

INDIA – Thermal Plants Installed Capacity



Capacity addition target and achievement for 2018-19:

For the year 2018-19, a capacity addition target of 8,106.15 MW was fixed. Against this, a capacity addition of 5921.755 MW was achieved. The Sector-wise / Fuel-wise break up is as under:

(In MW)

	Thermal		Hydro		Nuclear		Total	
	Target	Ach.	Target	Ach.	Target	Ach.	Target	Ach.
Central	2760	1960	710	110	-	-	3470	2070
State	4506.15	2849.755	130	30	-	-	4636.15	2879.755
Private	0	972	0	0	-	-	0	972
Total	7266.15	5781.755	840	140	-	-	8106.15	5921.755

Capacity addition target and achievement for 2019-20:

For the year 2019-20, a capacity addition target of 12,186.14 MW was fixed. Against this, a capacity addition of 7,065 MW was achieved. The Sector-wise / Fuel-wise break up is as under:

(In MW)

	Thermal		Hydro		Nuclear		Total	
	Target	Ach.	Target	Ach.	Target	Ach.	Target	Ach.
Central	6040	3940	600	300	700	-	7340.00	4240
State	4256.15	2780	210.99	-	-	-	4467.14	2780
Private	0	45	379	-	-	-	379.00	45
Total	10296.15	6765	1189.99	300	700	0	12186.14	7065

https://powermin.nic.in/sites/default/files/uploads/Capacity_Addition_Programme_Beyond_XII_Plan.pdf

INDIA – Thermal Plants Installed Capacity



Table 1 : Sector-wise breakup of installed capacity as on 31st August 2014

Sector	Installed capacity (MW)	% of total
Thermal (including gas and diesel)	176,118	69.5
Hydro	40,798	16.1
Nuclear	4,780	1.9
Renewable sources	31,692	12.5
Total	253,390	100%

Total Thermal	2,31,421	62.1%
Coal	1,99,595	53.6%
Lignite	6,360	1.7%
Gas	24,957	6.7%
Diesel	510	0.1%
Hydro (Renewable)	45,699	12.3%
Nuclear	6,780	1.8%
RES* (MNRE)	88,793	23.8%
Total	372,693	

* Installed capacity in respect of RES (MNRE) as on 31.08.2020.

<https://powermin.gov.in/en/content/power-sector-glance-all-india>

INDIA – Thermal Plants Installed Capacity

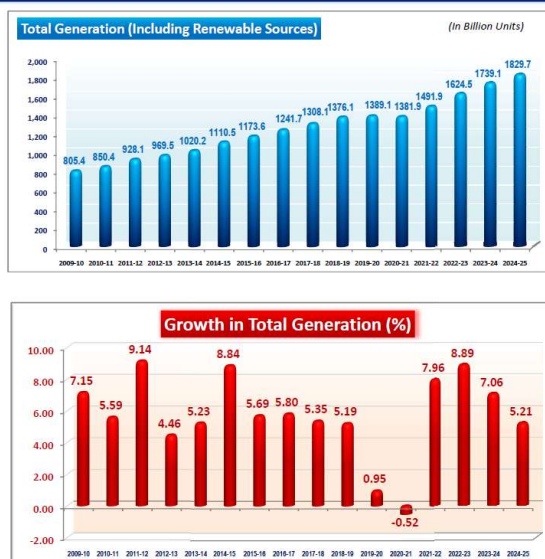


➤ Installed Generation Capacity (Sectorwise) as on 31.05.2025 :

Sector	Installed Capacity (MW)	% Share in Total
Central Sector	1,06,963	22.5%
State	1,10,417	23.2%
Private Sector	2,58,210	54.3%
TOTAL	4,75,590	100.0%

<https://powermin.gov.in/en/content/power-sector-glance-all-india>

INDIA – Thermal Plants Installed Capacity



<https://powermin.gov.in/en/content/power-sector-glance-all-india>

INDIA – Thermal Plants Installed Capacity



➤ Installed Generation Capacity (Fuelwise) as on 31.05.2025 :

Category		Installed Generation Capacity (MW)	% Share in Total
Fossil Fuel	Coal	2,12,718	44.7%
	Lignite	6,620	1.4%
	Gas	20,132	4.2%
	Diesel	589	0.1%
	Total Fossil Fuel :	2,40,060	50.5%
Non-Fossil Fuel	RES (Incl. Hydro)	2,26,751	47.7%
	Hydro	47,928	10.1%
	Wind, Solar & Other RE	1,78,823	37.6%
	Wind	51,292	10.8%
	Solar	1,10,834	23.3%
	BM Power/Cogen.	10,743	2.3%
	Waste to Energy	851	0.2%
	Small Hydro Power	5,102	1.1%
	Nuclear	8,780	1.8%
	Total Non-Fossil Fuel :	2,35,531	49.5%
Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)		4,75,590	100%

<https://powermin.gov.in/en/content/power-sector-glance-all-india>

Limitation of CCS at power stations



- Large-scale CCS deployment is unproven and decades away from being commercialized
- It is risky and expensive, and that a better option is renewable energy.
- Lack of accurate geological storage site data such as permeability, capacity etc.
- CCS technology leaves behind dangerous waste material that has to be stored, just like nuclear power stations.
- CCS technology is expected to use between 10% and 40% of the energy produced by a power station.
- Resource consumption will increase by one third with decrease in efficiency gains.
- Safe and permanent storage of CO₂ cannot be guaranteed. In 1986 a large leakage of naturally sequestered CO₂ rose from Lake Nyos in Cameroon and asphyxiated 1700 people.
- CCS has no effect on CO₂ emissions due to the mining and transport of coal.

CO₂ Abatement – Initiatives by GoI



- The **National Action Plan on Climate Change** (NAPCC) was launched in 2008 by the Prime Minister's Council on Climate Change.
- It includes various stake holders like public, agencies of government, scientists, industries, communities effected by climate change.
- There are 8 national missions forming the core of the NAPCC.
 - ✓ National Solar Mission
 - ✓ National Mission for Enhanced Energy Efficiency
 - ✓ National Mission on Sustainable Habitat
 - ✓ National Water Mission
 - ✓ National Mission for Sustaining the Himalayan Ecosystem
 - ✓ National Mission for Green India
 - ✓ National Mission for Sustainable Agriculture
 - ✓ National Mission on Strategic Knowledge for Climate Change

CO₂ Abatement – Initiatives by GoI



- **Retire old inefficient plants:** old units of coal-fired based plants need to be decommissioned. Already removed 5000 MW plants out of 15000 MW.
- **Renovation and Modernization of existing units:** Range of 200 to 500 MW plants need to be upgraded.
- **Adoption of supercritical technology:** Instead of 210/300/500/600 MW units, 660/800/1000 MW plants which have supercritical parameters.
- These will increase the efficiency around 1.5 to 2%.
- Introduction of **clean energy tax** on coal (imported and domestic)-**National clean energy fund**.
- **Renewable Energy Deployment – Wind, Solar, Biomass.**
 - ✓ Around increase of RES based energy from 12.5% (2014) to 23% (2020)
 - ✓ Cost, grid connectivity, peak load management, power tariffs are limitations that need to be overcome for deployment.
 - ✓ Sufficient technology should be developed in terms of materials, transmissions, distribution of power.

Measure of CO₂ Reduction - Initiatives by GoI



- **Renewable Energy Certificates:** 1 MWh of electricity generation from renewable sources of power gives a REC to producers. Power must be sold with last year average pricing of procurement.
- **PAT Scheme:** Perform, Achieve and Trade is market based mechanism to improve the energy efficiency.
- 1 tonne of oil equivalent (toe) energy savings will give 1 PAT certificate. REC and PAT certificates are interchangeable.

SECTORS

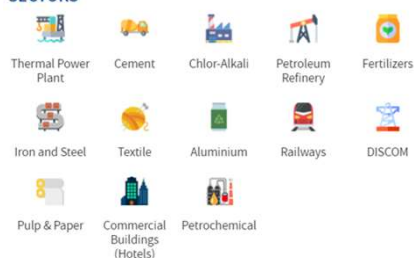


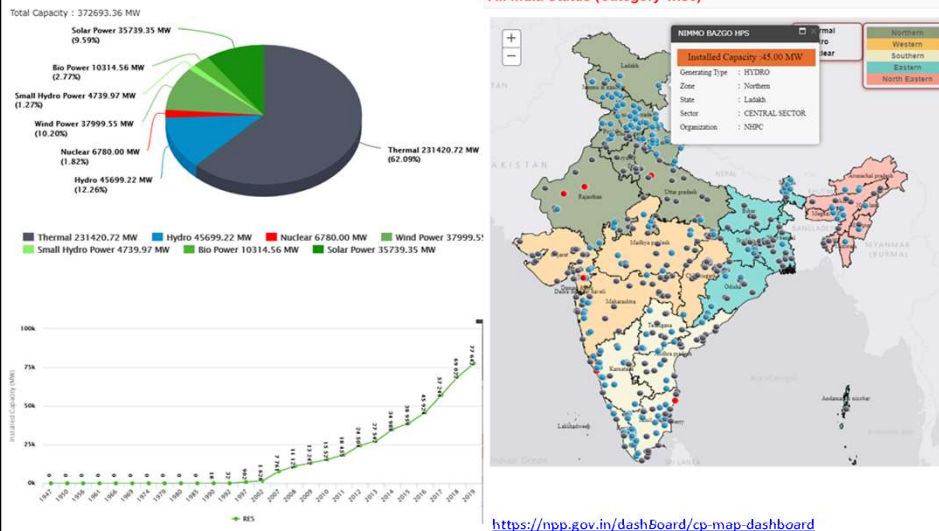
Table 3 : Designated consumers for energy conservation under PAT scheme

S.No.	Sector	No. of identified DCs	Annual energy consumption (Mtoe)	Share consumption (%)	Apportioned energy reduction for PAT Cycle-I (Mtoe)
1.	Power (Thermal)	144	104.56	63.38%	3.211
2.	Iron and steel	67	25.32	15.35%	1.486
3.	Cement	85	15.01	9.10%	0.815
4.	Aluminium	10	7.71	4.67%	0.456
5.	Fertilizers	29	8.20	4.97%	0.478
6.	Pulp and paper	31	2.09	1.27%	0.119
7.	Textile	90	1.20	0.73%	0.066
8.	Chlor-Alkali	22	0.88	0.53%	0.054
9.	Total	478	164.97	100.00%	6.686

Future Demand and Capacity Additions

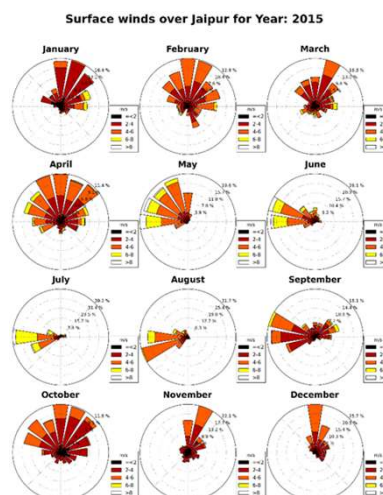
- Current energy demand is 380 GW which will project to 800 GW by 2040.

All India Status (Category-wise)



Future Demand and Capacity Additions

- Major challenges for RE integration:
 - Seasonal availability – Wind and Solar Energy
 - Wind/solar plants are generally located away from distribution center
- Renewable energy management centers with advanced weather monitoring system and forecasting tools.
- Smart grid: demand side management, energy conservation will optimize the resources.
- India Smart Grid Task Force - smart grids



<https://urbanemissions.info/india-apna/jaipur-india/>

Carbon Capture Storage and Utilization(CCSU)

- Industrializing carbon capture, storage, and utilization (CCSU) – a CO₂ sequestration technology.
- Removal from point source of excess CO₂ and utilizing or storing so as not to mix with atmosphere.
- Gas separation – absorption, membrane separation, adsorption, cryogenic separation
- Storage – sub-surface storage and utilization by producing energy fuels or minerals, industrial chemicals.
- Long distance between capture and storage – CO₂ transport in the form of liquid is needed.
- CCSU – cost depend on local fixing and deep storage conditions
- Post-combustion capture, In-combustion Processes, Pre-Combustion Technology, CO₂ Transportation and Injection Technologies, CO₂ Underground Geological Storage, CO₂ Fixation in the Terrestrial Ecosystem, CO₂ Utilization Technologies.

Carbon Capture Storage and Utilization(CCSU)

- **Post-combustion capture:** Involves Chemical, physical and cryogenic methods. Removing the CO₂ from flue gas at thermal power stations.
- Solvents such as amine based are used in chemical capture which are cost-intensive.
- Physical separation involves adsorption materials and membrane filters.
- Cryogenic separation involves cooling of flue gas components (gas to solid or liquid).
- **In-Combustion Processes:** supercritical combustion technology adaption will increase the power generation efficiency.
- This will increase concentration of CO₂ in flue gas which eventually reduce the cost while capture.
- Oxy-fuel combustion has limitation of oxygen separation costs from air.
- **Pre-Combustion Technology:** Proposed for capturing CO₂ from gasified or liquefied coal prior to combustion.
- Separation of H₂ and CH₄ from coal gas and IGCC are adopted.

Carbon Capture Storage and Utilization(CCSU)



- **CO₂ Transportation and Injection Technologies:** Compression is required for transportation to avoid huge volumes.
- For long range transportation – additional cost of 15 to 20%.
- Solid CO₂ is in ice form which can be stored or transported easily.
- Underground injection at suitable location is needed with technology.
- **CO₂ Underground Geological Storage:** Underground or in-depth of **saline waters, basalt rocks**, empty oil and gas fields, mineral rocks, under gas hydrates are proposed.
- Permanent CO₂ storage in sea beds, mineral rocks or gas hydrates is in research.
- **CO₂ Fixation in the Terrestrial Ecosystem:** Forests, trees, crops and soil – carbon sink.
- Terrestrial sequestration – biological or surface biosequestration, mining waste, marine algae sequestration.

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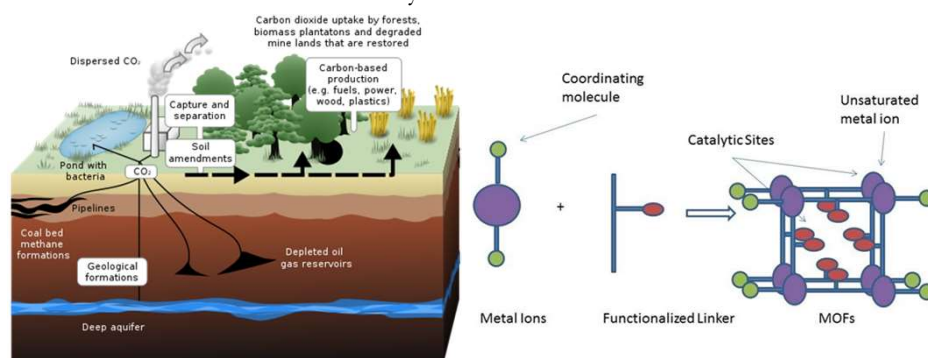
CH447 – CO₂ Capture, Sequestration & Utilization

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Carbon Capture Storage and Utilization(CCSU)



- **CO₂ Utilization Technologies:** Attractive, risk-free option into valuable products by chemical methods.
- Production of alcohols, fertilizers, carbonated drink feed stocks etc. – surface utilization
- Sub-surface utilization – injection into unminable coals will increase the coalbed methane or oil recovery.



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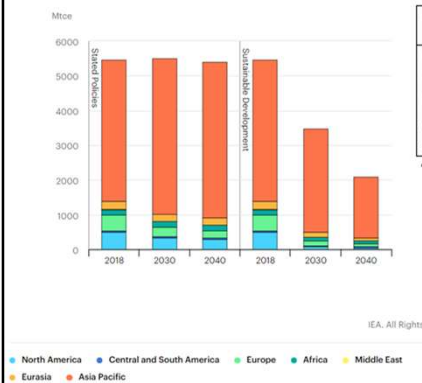
CH447 – CO₂ Capture, Sequestration & Utilization

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Global Coal Demand and Consumption



Coal demand by region and scenario, 2018-2040

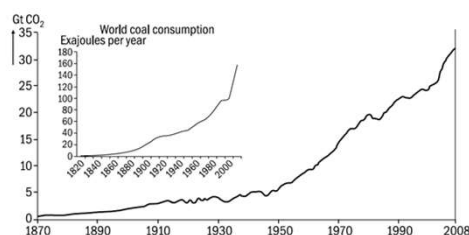


<https://www.iea.org/fuels-and-technologies/coal>

Table 1 : Increasing coal production and its current share in total energy for selected countries

Country	Coal production in 2006 in Mt	Coal production in 2012 in Mt	Share in total energy % in 2010
PRC	2,482.0	3,650.0	81
USA	990.0	935.0	50
India	427.0	595.0	68
Australia	309.0	431.2	69
South Africa	244.0	260.0	94

Source: World Coal Institute, 2013



✓ Asia Pacific region accounts for 67% of global production in 2011 as compared to 27% in 1981. This is projected to reach 78% by 2030.

Coal Reserves in INDIA



(Million tonnes)

State	Proved	Indicated	Inferred	Total
JHARKHAND	45563	31439	6150	83152
ODISHA	37391	34165	7739	79295
CHHATTISGARH	20428	34576	2202	57206
WEST BENGAL	14156	12869	4643	31667
MADHYA PRADESH	11958	12154	3875	27987
TELANGANA	10475	8576	2651	21702
MAHARASHTRA	7178	3074	2048	12299
ANDHRA PRADESH	0	1149	432	1581
BIHAR	161	813	392	1367
UTTAR PRADESH	884	178	0	1062
MEGHALAYA	89	17	471	576
ASSAM	465	57	3	525
NAGALAND	9	0	402	410
SIKKIM	0	58	43	101
ARUNACHAL PRADESH	31	40	19	90
Total	148787	139164	31069	319020

(Source: Geological Survey of India)

<https://coal.nic.in/content/coal-reserves>

Coal Reserves in INDIA



(Million tonnes)

Formation	Proved/Measured	Indicated	Inferred	Total
Gondwana Coals	148194	139065	30174	317433
Tertiary Coals	594	99	895	1588
Grand Total	148787	139164	31069	319020

TYPE AND CATEGORY-WISE COAL RESOURCES OF INDIA

The Type and Category-wise Coal resources of India as on 1.4.2018 are given in table below:

(Million tonnes)

Type of Coal	Proved/Measured	Indicated	Inferred	Total
(A) Coking :-				
-Prime Coking	4649	664	0	5313
-Medium Coking	13914	11709	1879	27502
-Semi-Coking	519	995	193	1708
Sub-Total Coking	19082	Click here for English version	073	34522
(B) Non-Coking :-	129112	125697	28102	282910
(C) Tertiary Coal :-	594	99	895	1588
Grand Total	148787	139164	31069	319020

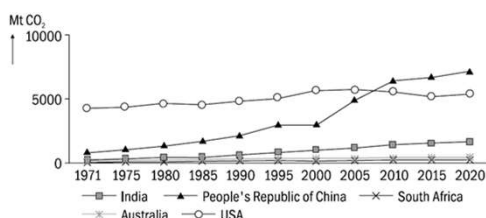
<https://coal.nic.in/content/coal-reserves>

Commitment of Different Countries



Table 2 : Coal-fired plants in 2010 [17]

Country	No. of plants	Total installed capacity in GW
PRC	2,928	669.2
USA	1,368	336.3
India	809	100.5
South Africa	114	37.5
Australia	109	29.97



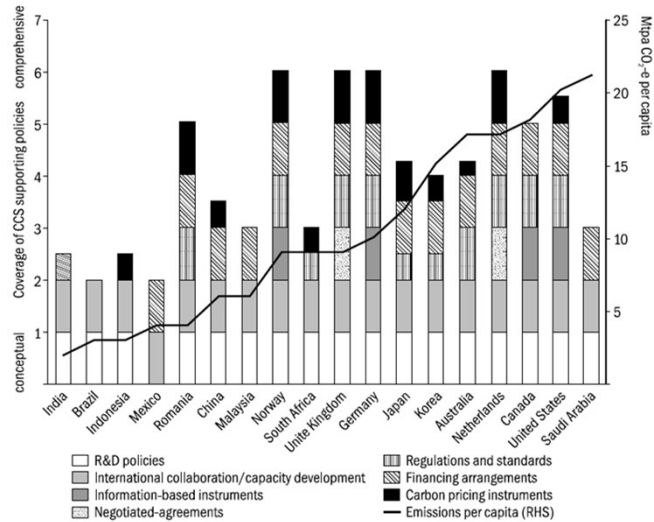
It is predicted that the average global temperature could increase between 1.1–6.4°C by 2100.

Table 3 : CO₂ emission reduction commitment of selected countries

Country	GDP per capita in USD (2011)	Emission reduction target and pledges				Date of ratification	
		By 2020 (Unconditional)	By 2020 (Conditional)	By 2050 (Conditional)	Other	UNFCCC	Kyoto Protocol
Australia	67,039	-5% relative to 2000	Up to -15% relative to or - 25% relative to 2000	-80% relative to 2000		30 Dec 1992	12 Dec 2007
USA	49,922	-17% by 2020 relative to 2005		Towards a goal of - 83% relative to 2005	-30% in 2025, -42% in 2030 relative to 2005	15 Oct 1992	-
South Africa	8,090		-34% relative to BAU	-42% by 2025 relative to 2005 and missions to peak between 2020 and 2025		29 Aug 1997	31 Jul 2002
PRC	5,439		-40 to -45% of CO ₂ per unit of GDP to 2005	-17% of CO ₂ per unit of GDP by 2015 relative to 2005		5 Jan 1993	30 Aug 2002
India	1,528		-20% to -5% emission intensity per unit of GDP relative to 2005			1 Nov 1993	26 Aug 2002

Source: Compiled using data from various sources including <http://unstats.un.org>

CCSU – Current Technology and Policy Status

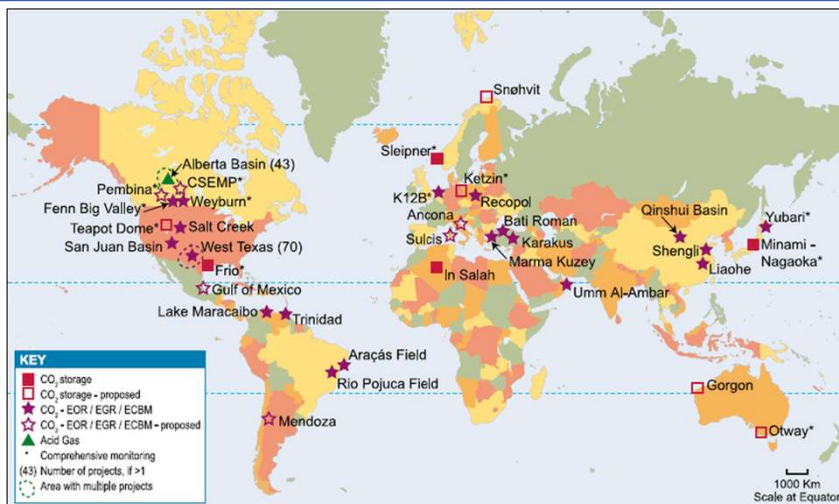


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CCSU – Major Global Projects



Messner, Steve & Robertson, Kate & Findsen, Jette. (2006). International Carbon Capture and Storage Projects Overcoming Legal Barriers, DOE/NETL-2006/1236.

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CCSU – Current Technology and Policy Status



- **Australia:** Rich in natural resources like coal, petroleum, natural gas. About 70% electricity is produced from coal based plants.
- Australia has carbon tax on coal. Invested 1.9 billion \$ to build 2 to 4 CCS projects on 1000 MW plants.
- **Unites States of America:** Second largest emitter of CO₂ and biggest economy.
- Shale gas production has reduced coal consumption by 1.5% in 2012. It has target of 2020 to reduce the emissions by 17% with a reference of 2005 base.
- US is having 113 CCS projects out of a total 240 across the world. These consists of 24 large scale projects as well.
- USA is developing a legal framework for CCS and its deployment.
- **South Africa:** coal based electricity is 93% with a world leader in liquefaction of coal technology.
- Carbon tax, no to new coal based plants, CCS application are in force.

CCSU – Current Technology and Policy Status

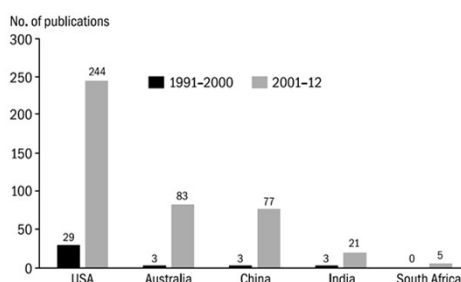


- **China:** Coal reserves of 114500 billion tonnes(~14% of world) with an installed capacity of 700 GW coal based plants.
- A total of 11 CCS projects are in development. The reduction targets are around 40 – 45 % of 2005 emission baseline by 2020.
- International cooperation with EU, UK and Canada – Near Zero Emissions Cola Initiative.
- **India:** Third largest producer and consumer of coal. Electricity generation share of coal is 75%. Early policies focused on the energy conservation, adaption to climate change, renewable energy introduction.
- Introduction of **PAT** and **RES** certificates, targets for nine industrial sectors to improve the efficiency.
- DST enables R&D aim to control GHG emissions. Voluntary reduction of 20 – 25% of CO₂ emissions with reference to 2005 emissions.
- National Action Plan for Climate Change – 8 National missions.
- Setup of National Clean Energy Fund (NCEF) – clean energy technology.

CCSU – Current Technology and Policy Status



- Deployment of CCS is not a India policy - No legislation on CCSU.
- Technology of CCS is not **fully developed** to the commercial scale.
- **High investment** is needed. Almost 100 billion dollars per annum in CCS adoption to thermal / other industrial plants.
- CCS will increase the price of electricity which will be a big burden to domestic and industrial users.
- Availability of suitable resources and risks/ safety in underground storage.



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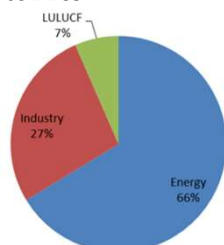
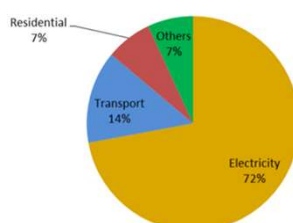
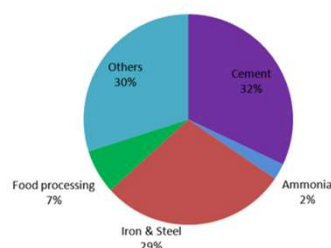
CH447 – CO₂ Capture, Sequestration & Utilization

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- TERI Conducted a study for CCS in India along with Global Carbon Capture and Storage Institute (GCCSI).
- The study focused on the issues – opportunities and barriers for the CCS deployment and implementation in India.
- India per-capita emission – 1.3 tonnes with 1.7Gt of CO₂ in 2007.
- **IEP** – Integrated energy policy, India CO₂ generation can be in the range of 4 – 5.5 Bt by 2030. The percapita emissions will reach around 2.6 – 3.6 tonnes.

2007 - CO₂ emissionsCO₂ emissions break-up

<https://www.globalccsinstitute.com/archive/hub/publications/88981/india-ccs-scoping-study-final-report.pdf>

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Table 2: List of UMPPs awarded [27]

Name of UMPP	Type	Special purpose vehicle	Developer awarded	Current status
Mundra, Gujarat (4000 MW)	Coastal	Coastal Gujarat Power Ltd.	Tata Power Ltd.	4th Unit (each unit of 800 MW) commissioned in Jan 2013 [28]
Sasan, Madhya Pradesh (3960 MW)	Pithead	Sasan Power Ltd.	Reliance Power Ltd.	1st Unit (660 MW) to be commissioned in Jan 2013 [29]
Krishnapatnam, Andhra Pradesh (3960 MW)	Coastal	Coastal Andhra Power Ltd.	Reliance Power Ltd.	Work stopped due to change in Indonesian coal regulations [29]. Central Electricity Regulatory Commission (CERC) approached for tariff hike [30]
Tilaiya, Jharkhand (3960 MW)	Pithead	Jharkhand Integrated Power Ltd.	Reliance Power Ltd.	Land acquisition and financing in progress [31]

Table 3: List of UMPPs for which SPVs incorporated [26]

Name of UMPP	Type	Special Purpose Vehicle
Chhattisgarh	Pithead	Chhattisgarh Sarguja Power Ltd.
Orissa	Pithead	Orissa Integrated Power Ltd.
Cheyur, Tamil Nadu	Coastal	Coastal Tamil Nadu Power Ltd.
Andhra Pradesh 2 nd UMPP	Coastal	Tatiya Andhra Mega Power Ltd.
Orissa Additional UMPP 1	Pithead	Sakhigopal Integrated Power Co. Ltd.
Orissa Additional UMPP 2	Pithead	Ghogarpalli Integrated Power Co. Ltd.
Maharashtra UMPP	Coastal	Coastal Maharashtra Power Ltd.
Karnataka UMPP	Coastal	Coastal Karnataka Power Ltd.

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- Levelised Cost of Electricity (LCOE, Rs/kWh) – parameter to measure how much amount of money is invested per unit of power production.

$$\text{LCOE} = (C_{\text{cap}} + C_{\text{O\&M}}) * a / h + C_{\text{TS}} + C_{\text{fuel}}$$

Where C_{cap} = Specific capital cost (Rs/kW)

$C_{\text{O\&M}}$ = Specific operating and maintenance cost (Rs/kW)

a = annuity factor, as calculated by the formula

$$a = I * (1+I)^n / ((1+I)^n - 1)$$

I = Interest rate (%)

n = Depreciation period (years)

C_{TS} = Specific transport and storage cost (Rs/kWh)

C_{fuel} = Specific fuel cost (Rs/kWh)

h = Annual hours of operation

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Table 7: Assumptions and technical specifications considered for the base case power plant

Item	Value	Units	Explanation/reference
Plant type	Supercritical, capture ready		
Plant size	4000	MW	CCS implementation is expected to be more cost effective for larger power plants, and 4000 MW plants are among the largest presently in operation. This plant is assumed to have 5 units of 800 MW each.
Plant life	40	Years	[85]
Plant start year	2015	-	
CCS start year	2020	-	Since CCS technology is expected to take a few years to mature, it is postulated that the carbon capture will be retrofitted to the plant after 5 years of operation.
Location	Indian coast for imported coal/pithead for Indian coal		Most large scale power plants are being planned either near a captive coal mine or along the coast where coal import is possible.
PLF	90%		-
Net power	3720	MW	Assuming auxiliary power consumption of 7%, as given in [87].
CCS operational life	35	Years	-
Specific CO ₂ emissions	1	kg/kWh	As given in Section 3, specific emissions from Indian coal fired plants are 1.07 kg/kWh. Being more efficient than conventional plants, emissions from supercritical units are expected to be lower.
Plant heat rate	8899.2	MJ/MWh	Net heat rate taken from [88]
Calorific value of coal used	5350	kCal/kg	Gross calorific value taken from [88]
Fuel cost	120	USD/t	For imported coal, as per [89], with a transport cost of 39 \$/t as per [90]
	65	USD/t	For Indian coal, as per [91], with a transport cost of 39 \$/t as per [90]

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Table 8: Assumptions related to carbon dioxide capture, transport and storage

Item	Value	Units	Explanation/reference
Gross power output	4800	MW	Due to addition of an 800 MW unit.
Net power output	3930	MW	Based on auxiliary energy consumption data from [85].
Heat rate for retrofitted plant	11836	MJ/MWh	For 33% capture penalty.
CO ₂ capture rate	90%		[85]
Capture technology	Monoethanolamine (MEA) based post combustion capture		MEA based capture is the most developed method of carbon capture, and post combustion capture is the most suitable for retrofitting.
Amount of CO ₂ captured/stored	34	MT per annum	-
Pipeline specifications	42" pipeline on-shore, with booster stations every 100km, 24" pipelines offshore		[85]
Pipeline onshore distance	200	Km	-
Pipeline offshore distance	50	Km	-
Storage site	Saline aquifers		Although storage data is in general uncertain for India, [93] indicates that saline aquifer potential may be the highest, and hence has been considered.
Injection specifications	12 injection wells each with 8000 t CO ₂ /day capacity		Modified from [85] based on higher storage requirements for this study.

[port.pdf](#)

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Table 9: Cost figures considered in Case 1

Item	Value	Units	Explanation/reference
Base plant capital cost	4000	USD mn	[27]
Capture capital cost	2438	USD mn	Value given in [85] adjusted for the larger unit size and lower base plant capital cost in present case.
Transport capital cost	90	USD mn	Adapted from [85] based on the pipeline lengths considered as per Table 8. Since [85] indicates overall capital costs for transport and storage, the ratio of capital costs mentioned in [94] has been used to split the costs.
Storage capital cost	170	USD mn	Storage capital cost arrived at in the same way as the transport capital cost, and then increased by 20% to account for the larger injection rate in the present study as compared to [85].
Monitoring capital cost	26	USD mn	Costs given in [95] extrapolated to injection levels of present study.
Fixed O&M for base plant	72.45	USD mn/y	[87]
Variable O&M for base plant	3.40	USD/MWh	[87]
Fixed O&M for capture equipment retrofitted plant	94.19	USD mn/y	Increase in O&M costs taken to be 30%, in line with [85]
Variable O&M for capture equipment retrofitted plant	4.42	USD/MWh	Increase in O&M costs taken to be 30%, in line with [85]
Operating cost for transport and storage	2	USD mn/y	Taken as average of costs for coastal power plants as listed in [85].
Monitoring operating cost	154	USD mn	Calculated similar to monitoring capital cost.

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Table 10: Case 1 financial assumptions

Item	Value	Units	Explanation/Reference
Debt: equity ratio	80 : 20		
Loan amount	3200	USD mn	
Repayment period	15	Years	
Interest rate	7.05%		[96]
Exchange rate	1 USD = 50 INR		

Table 11: LCOE for Case 1

Item	Imported coal	Indian coal
LCOE without capture (Rs/kWh)	2.95	1.85
LCOE with capture (Rs/kWh)	4.02	2.56
LCOE with CCS (Rs/kWh)	4.04	2.59
LCOE with CCS and monitoring (Rs/kWh)	4.06	2.61
Increase in LCOE due to CCS and monitoring	38%	41%

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Table 12: Differences between Case 1 and Case 2, with explanations

Item	Case 1	Case 2	Unit	Explanation
Base plant capital costs	4000	4800	USD mn	Stakeholders stated that the capital cost, while site-specific, should in general be around USD 1.2 million/MW.
Capture capital costs	2438	3005	USD mn	Stakeholders pointed out that the capture capital cost should not be a function of the base plant capital cost, and instead absolute values should be taken. Since the revised base plant capital was similar to that in [85], the capital cost of CCS equipment given there was adopted, with adjustment for the larger unit size.
Plant load factor	90%	85%		As per CERC norms [97].
Plant life	40	25	years	As per stakeholder consultation.
Base plant net heat rate	8899.2	10136	MJ/MWh	CERC norms state a heat rate of 2425 kCal/kWh [97], and this value has been used for Case 2. This figure was also used to recalculate the gross heat rate after capture assuming the same 33% capture penalty as earlier.
Interest rate	7.05%	13%		According to a stakeholder, the debt interest rate taken in Case 1 was unrealistically low, and a figure of around 13% was suggested.
Debt equity ratio	80/20	70/30		As per tariff policy, Government of India [98].
Calorific value of Indian coal	5350	3500	kCal/kg	According to a stakeholder, from their experience, the calorific value of the coal available domestically at 65 USD/t is around 3500 kCal/kg.

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Table 13: LCOE for Case 2

Item	Imported coal	Indian coal
LCOE without capture (Rs/kWh)	3.97	3.50
LCOE with capture (Rs/kWh)	5.52	4.90
LCOE with CCS (Rs/kWh)	5.58	4.95
LCOE with CCS and monitoring (Rs/kWh)	5.61	4.99
Increase in LCOE due to CCS and monitoring	41 %	42 %

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Table 14: Differences between Case 2 and Case 3, with explanations

Item	Case 2	Case 3	Unit	Explanation
Capture capital costs	3005	3350	USD mn	The capture equipment cost in [86] for a base plant size of 580 MW is USD 418 mn, and capture material cost is USD 15 mn. This has been extrapolated to a plant size of 3960 MW using an extrapolation factor of 0.8, which is higher than the 0.6 factor generally used for scaling up economics, due to CCS technology not being demonstrated at a large scale, and hence possibly not scaling up as smoothly as established technologies. The location specific factors mentioned in [86] have been used to estimate costs for an Indian scenario. The cost of the extra unit has been taken by dividing the capital cost of the base plant by the number of units (6).
Plant size	4000	3960	MW	In this case, a power plant with 6 units of 660 MW, with the addition of another 660 MW unit to compensate for the capture penalty, was considered. This is consistent with changes in UMPT norms as mentioned in [28].
Gross power output with CCS	4800	4620	MW	Same as above.
Net power output without CCS	3720	3755	MW	The auxiliary power consumption considered is lower in [86], even for the base power plant without capture equipment installed.
Net power output with CCS	3930	3805	MW	The auxiliary power consumption with CCS considered is lower in [86].
Base plant net heat rate	9000	9200	MJ/MWh	As given in [86].
Capture equipment net heat rate	11970	13220	MJ/MWh	As given in [86].
Calorific value of Indian coal	3500	3014	kCal/kg	As given in [86].
Cost of Indian coal	65	56	USD/t	As calculated from [86].
Amount of CO ₂ to be captured/stored	34	32.8	MTPA	Reduction in amount of CO ₂ captured occurs due to reduction in plant size.

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Table 15: Cost of electricity for Case 3

Item	Imported coal	Indian coal
LCOE without capture (Rs/kWh)	3.73	3.31
LCOE with capture (Rs/kWh)	5.39	4.78
LCOE with CCS (Rs/kWh)	5.44	4.84
LCOE with CCS and monitoring (Rs/kWh)	5.48	4.87
Increase in LCOE due to CCS and monitoring	47%	47%

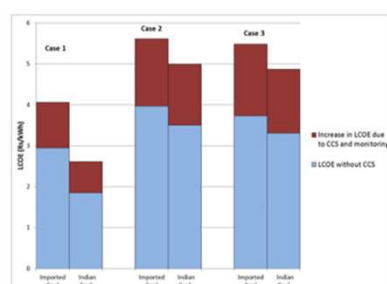


Figure 10: Comparison of the Levelised Cost of Electricity (LCOE) and the increase in LCOE for the different cases

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- ✓ From these three cases, it can be concluded that the increase in the LCOE is roughly 38-47%.
- ✓ However, very large plant is studied for CCS implementation.

CCS – India Status

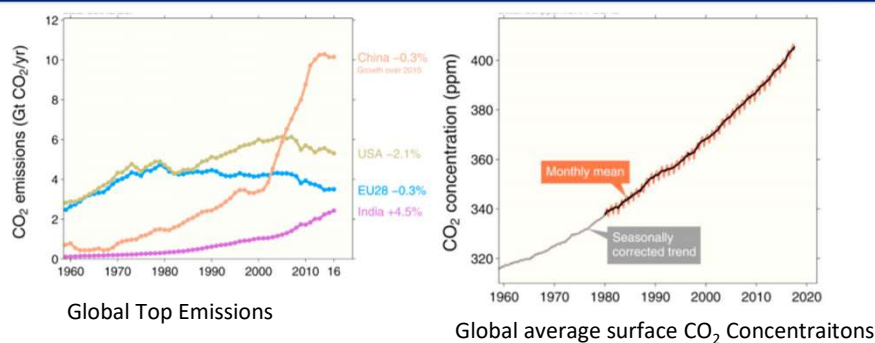


Table 1: List of Commercial C2 storage plants in India [15]

Plant	Main Developer	Capture technology	Industry	CO ₂ absorption capacity in Tonnes per day (TPD)
Aonla (U.P.) urea plant	Indian Farmers Fertilizer Co-Operative	Amine Based Combustion Capture	(Post- Chemical Production)	450 TPD
Jagdipshpur (Orissa) urea plant	Indo Gulf Corporation	Amine Based Combustion Capture	(Post- Chemical Production)	150 TPD
Phulpur (U.P.) urea plant	Indian Farmers Fertilizer Co-Operative	Amine Based Combustion Capture	(Post- Chemical Production)	450 TPD

Abhishek Gupta et al. Carbon capture and sequestration potential in India: A Comprehensive review Energy Procedia 160 (2019) 848–855

CCS – Global Status

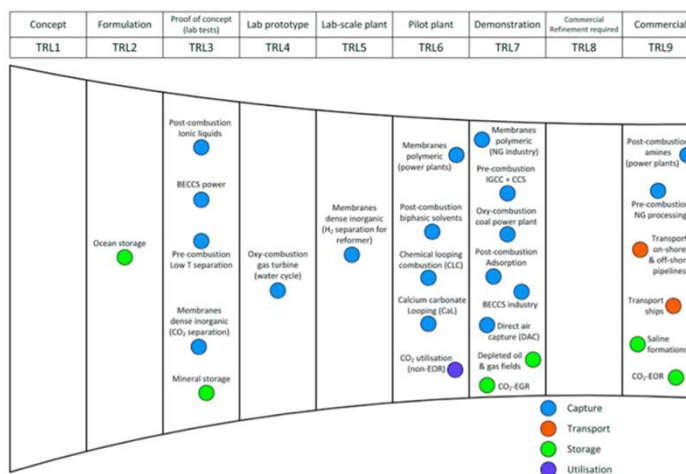


Fig. 4 Current development progress of carbon capture, storage and utilization technologies in terms of technology readiness level (TRL). BECCS = bio-energy with CCS, IGCC = integrated gasification combined cycle, EGR = enhanced gas recovery, EOR = enhanced oil recovery, NG = natural gas. Note: CO₂ utilization (non-EOR) reflects a wide range of technologies, most of which have been demonstrated conceptually at the lab scale. The list of technologies is not intended to be exhaustive [12].

Abhishek Gupta et al. Carbon capture and sequestration potential in India: A Comprehensive review Energy Procedia 160 (2019) 848–855

Limitation of CCS in INDIA



- Large-scale CCS deployment is unproven and decades away from being commercialized
- It is risky and expensive, and that a better option is renewable energy.
- Lack of accurate geological storage site data such as permeability, capacity etc.
- CCS technology leaves behind dangerous waste material that has to be stored, just like nuclear power stations.
- CCS technology is expected to use between 10% and 40% of the energy produced by a power station.
- Resource consumption will increase by one third with decrease in efficiency gains.
- Safe and permanent storage of CO₂ cannot be guaranteed. In 1986 a large leakage of naturally sequestered CO₂ rose from Lake Nyos in Cameroon and asphyxiated 1700 people.
- CCS has no effect on CO₂ emissions due to the mining and transport of coal.

Successful Roadmap for CCS in INDIA



- ❖ Policy and regulation framework
 - ✓ Private finance for demonstration and early deployment of CCS
 - ✓ National law amendments, regulations and guidelines review
- ❖ Identification of suitable CO₂ storage
 - ✓ Implementation of policies for exploration, characterization, frameworks for safe CCS
- ❖ Improvement and Cost reduction of Capture technologies
 - ✓ R&D in novel capture technologies and power generation cycles
 - ✓ International collaboration
- ❖ Development of CO₂ Transport Infrastructure
 - ✓ Pipelines, shipping for CO₂
 - ✓ Efficient transport will reduce the cost

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